Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering (SBASSE)
AI651 – Deep Learning for Space, Time and Graphs
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Assignment 2 – Report

Name: Hasan Amjad **Roll No.:** 25280042
GitHub repository: [TODO: add public repo link]

Summary. This report presents the work done for both questions of AI651 Assignment 2. Question 1 proposes a forecasting system that issues one-hour-ahead water-level forecasts for stations S1–S4 every ten minutes. The design places all sources on one space–time grid, summarises each location's history with a causal temporal convolution, and passes information downstream along the river with attention that learns the travel delay. Question 2 [TODO: one-line summary of Q2 once results are in].

Question 1 – River Forecasting Proposal

Service. Every ten minutes, forecast the water level at the four main-river stations S1–S4 for each ten-minute step of the next hour.

Design in one sentence. All sources are placed on one space–time grid (places × ten-minute steps). A causal temporal convolution summarises what has happened at each place. Four rounds of *upstream attention* then pass information down the river, with the travel delay learned rather than fixed. A small head predicts each station's change in level over the next hour.

1. Architecture diagram

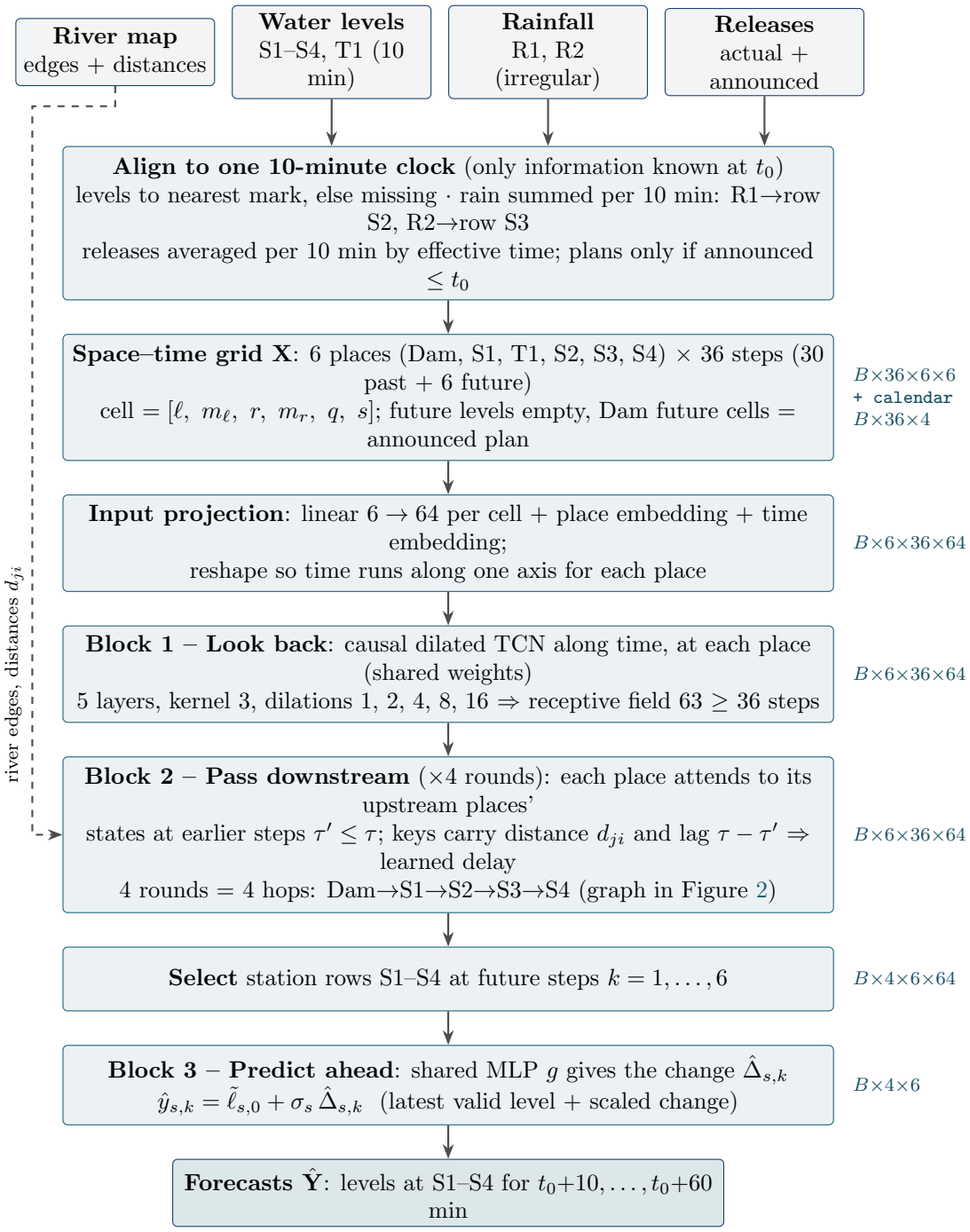


Figure 1: Proposed forecasting system. Grey boxes are data sources; teal boxes are processing blocks, with tensor shapes on the right (B = issuances in a batch).

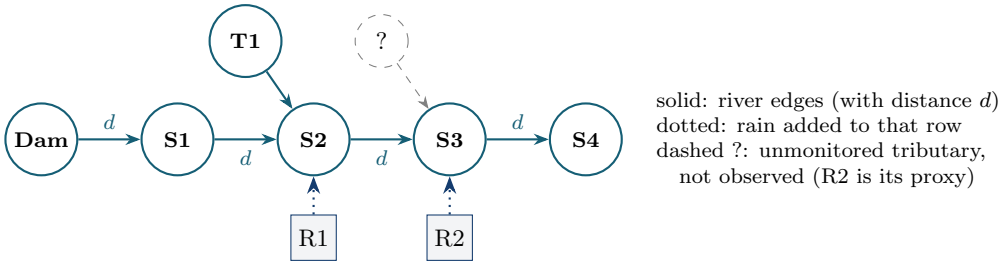


Figure 2: River graph used by Block 2. Edges point downstream; "upstream of i" means the places with an edge into i.

2. Assumptions

Supplied facts (from the handout), kept separate from assumptions. F1 levels at S1–S4 and T1 every 10 minutes; F2 rainfall at R1, R2 with timestamps and gauge locations; F3 recorded releases, and future release schedules with announcement and effective times; F4 station locations, connectivity and river distances; F5 a monitored tributary (T1) joins between S1 and S2, an unmonitored one between S2 and S3; F6 stations are unevenly spaced and travel times change with conditions; F7 future rainfall, future levels, exact travel times and the unmonitored inflow are not available.

Table 1: Assumptions, why each is reasonable, and its effect on the design.

	Assumption	Why it is reasonable	Effect on the design
Data conditions the handout leaves unspecified			
A1	At least 3 years of 10-minute records exist, including several high-water events.	Telemetry networks of this kind usually keep multi-year archives.	Chronological train/validation/test split by year; a small model (≈0.2M parameters) to suit limited flood examples.
A2	All timestamps use one clock (e.g. UTC); level readings are instantaneous values at their timestamp.	Standard for hydrometric telemetry.	Levels snapped to the nearest 10-minute mark (±2 min), otherwise marked missing.
A3	Each rain reading is the total since the previous reading (tipping-bucket style), at irregular times.	The handout gives timestamps but not the reading's meaning; this is the common gauge output.	Rain summed into 10-minute bins by observation time.
A4	Levels normally arrive within ∼5 min, rain within ∼20 min, releases within 10 min; occasionally hours late.	Typical of radio/cellular telemetry.	Forecast issued on time with whatever has arrived; late cells treated as missing (Section 5).
A5	Data contain gaps, spikes and flat-lines, but no systematic long-term drift.	Common sensor faults.	Simple quality screen marks impossible jumps as missing; a missing-value flag per cell.
A6	Release schedules are archived with announcement and effective times, so the plan known at any past t_0 can be rebuilt.	The authority records announcements (F3).	Training uses the plan *as announced*, not the actual release, so there is no leakage.
A7	Water travels from the dam to S4 in at most ∼4 hours at low flow, and rainfall reaches the river mostly within ∼5 hours.	The illustration suggests a short mountain river; to be checked on data.	History window $H = 30$ steps (5 h).

	Assumption	Why it is reasonable	Effect on the design
A8	Flood alert levels θ_s for each station are known, or can be set at a high percentile of the record.	Authorities normally define warning levels.	Higher loss weight above θ_s.
River dynamics assumed by the model			
A9	Water only moves downstream; backwater from junctions does not affect upstream gauges within an hour.	True for a steep river with free-flowing junctions.	Information passes only along downstream edges.
A10	Travel time is positive, varies with flow (faster at high water), and changes smoothly.	Wave speed rises with depth (F6).	Delay is learned by attention that depends on the current state; not hard-coded.
A11	R1 represents rain on the local catchment between S1 and S2; R2 represents the unmonitored tributary's catchment, which drains to S3.	Gauge positions in the handout figure.	R1 is placed in row S2 and R2 in row S3.
A12	The unmonitored inflow is driven mainly by rain on its catchment, plus a slow baseflow.	Usual behaviour of small tributaries.	R2 history is a usable proxy; the remaining error appears as wider uncertainty at S3/S4.
A13	Releases are the only controlled inflow; withdrawals between stations are negligible.	No abstractions are mentioned.	No extra input rows needed.
A14	Without a newer announcement, the current release continues.	Operators usually change releases by announcement.	Future Dam cells filled with the current rate when no plan exists (flag $s = 1$).
A15	Gauge datums and channel shape are stable over the record.	No changes are mentioned.	One model over the whole record; retrain if a datum changes.

3. Forecasting task and data specification

Issuance and horizon. A forecast is issued every 10 minutes at time t_0 (on the :00, :10, ... marks). It predicts the level at S1–S4 at the six lead times $t_0 + 10k$ minutes, $k = 1, \ldots, 6$, so the one-hour horizon is covered at ten-minute resolution: a 4×6 output per issuance.

History window. The model sees the last $H = 30$ steps (5 hours, $t_0 - 290$ min to t_0). This covers the slowest assumed travel and rainfall response times (A7), with margin. The actual delay inside the window is learned.

Grid. All inputs share one grid with $T = H + 6 = 36$ columns ($\tau = -29, \ldots, 0$ past; $\tau = 1, \ldots, 6$ future) and $P = 6$ rows (Dam, S1, T1, S2, S3, S4). Each cell holds $F = 6$ features: level ℓ, level-observed flag m_ℓ, rain r, rain-observed flag m_r, release q and planned-release flag s. The two extra flags separate "no rain" from "rain unknown", and "planned" from "measured" release (needed in Section 5).

Table 2: Inputs and outputs for one batch. B = number of issuances (forecast times) in the batch.

Item	Shape	Axes	Time coverage
Grid $\mathbf{X}$	$B \times 36 \times 6 \times 6$	issuance, column τ, place, feature	$t_0 - 290$ min … $t_0 + 60$ min
level ℓ, flag m_ℓ	(in $\mathbf{X}$)	rows S1–S4, T1	$\tau \le 0$ as observed; $\tau > 0$ empty, $m_\ell = 0$
rain r, flag m_r	(in $\mathbf{X}$)	R1 in row S2, R2 in row S3	$\tau \le 0$; $\tau > 0$ empty, $m_r = 0$
release q, flag s	(in $\mathbf{X}$)	Dam row	$\tau \le 0$ actual ($s = 0$); $\tau > 0$ plan announced by t_0 ($s = 1$)
Calendar $\mathbf{C}$	$B \times 36 \times 4$	issuance, column, sin / cos of time of day and day of year	same as $\mathbf{X}$
River map $\mathcal{E}$	5 edges, each with distance d_{ji}	Dam→S1, S1→S2, T1→S2, S2→S3, S3→S4	fixed
Output $\hat{\mathbf{Y}}$	$B \times 4 \times 6$	issuance, station S1–S4, lead time k	$t_0 + 10$ … $t_0 + 60$ min
Target $\mathbf{Y}$, mask $\mathbf{M}$	$B \times 4 \times 6$	as $\hat{\mathbf{Y}}$ (training only)	same

Alignment across sampling times. All sources use one ten-minute clock. A level reading goes in column τ if its timestamp is within ± 2 minutes of $t_0 + 10\tau$; otherwise the cell is missing. Rain totals (A3) are summed into the interval $(t_0 + 10(\tau - 1),\ t_0 + 10\tau]$ by observation time. Releases are averaged over each interval and placed by their *effective* time.

Alignment across locations. Values are not interpolated in space. Each source is assigned to one place row (rain via A11), and the river distances enter as features on the edges of the graph used in block 2. Unequal spacing is therefore handled through distances and learned delays, not by treating rows as evenly spaced neighbours.

Information available at issuance. A cell is filled only if its value is known at t_0: a measurement must have *arrived* by t_0, and a release plan must have been *announced* by t_0. Future levels and future rainfall are never available; their cells stay empty with flags set to 0. The only filled future cells are the Dam row's announced releases.

4. Model computations and design justification

Notation. Places $p \in \{\text{Dam}, \text{S1}, \text{T1}, \text{S2}, \text{S3}, \text{S4}\}$; columns $\tau \in \{-29, \dots, 6\}$; stations $s \in \{\text{S1}, \dots, \text{S4}\}$; lead times $k = 1, \dots, 6$; hidden width $d = 64$. $\mathbf{x}_{p,\tau} \in \mathbb{R}^6$ is one grid cell, $\mathcal{U}(i)$ the set of places directly upstream of i (from $\mathcal{E}$), and d_{ji} the river distance from j to i. Levels are standardised per station using training statistics.

Input projection.

$$\mathbf{z}^{(0)}_{p,\tau} = W_{\text{in}}\, \mathbf{x}_{p,\tau} + \mathbf{e}_p + \mathbf{c}_\tau, \qquad \mathbf{z} \in \mathbb{R}^d, \tag{1}$$

where $\mathbf{e}_p$ is a learned place embedding and $\mathbf{c}_\tau$ a time embedding (calendar features plus column position, so future columns are distinguishable).

Block 1 – Look back (causal dilated TCN, applied to each place with shared weights). For layers $l = 1, \dots, 5$ with kernel size 3 and dilation $\delta_l = 2^{l-1}$:

$$\mathbf{z}^{(l)}_{p,\tau} = \mathbf{z}^{(l-1)}_{p,\tau} + \text{ReLU}\Big(\textstyle\sum_{a=0}^{2} A^{(l)}_a\, \mathbf{z}^{(l-1)}_{p,\,\tau - a\delta_l} + \mathbf{b}^{(l)} \Big), \tag{2}$$

with zero padding before $\tau = -29$. Only columns $\le \tau$ are used (causal), and the receptive field is $1 + 2(1 + 2 + 4 + 8 + 16) = 63 \ge 36$ columns, so every column can see the whole window. Set $\mathbf{h}^0_{p,\tau} = \mathbf{z}^{(5)}_{p,\tau}$.

Block 2 – Pass downstream (upstream lag attention, $R = 4$ rounds, separate weights per round). For round r and each place i with $\mathcal{U}(i) \neq \emptyset$:

$$\mathbf{q}_{i,\tau} = W_Q^{(r)} \mathbf{h}_{i,\tau}^{r-1}, \qquad \mathbf{k}_{j,\tau'} = W_K^{(r)} \mathbf{h}_{j,\tau'}^{r-1} + \mathbf{u}_{\tau-\tau'}^{(r)} + \mathbf{w}^{(r)} d_{ji}, \qquad \mathbf{v}_{j,\tau'} = W_V^{(r)} \mathbf{h}_{j,\tau'}^{r-1}, \tag{3}$$

$$\alpha_{i\tau,\, j\tau'} = \underset{j \in \mathcal{U}(i),\ \tau' \leq \tau}{\operatorname{softmax}} \Big(\mathbf{q}_{i,\tau}^{\top} \mathbf{k}_{j,\tau'} / \sqrt{d} \Big), \qquad \mathbf{m}_{i,\tau} = \textstyle\sum_{j,\tau'} \alpha_{i\tau,\, j\tau'}\, \mathbf{v}_{j,\tau'}, \tag{4}$$

$$\tilde{\mathbf{h}}_{i,\tau} = \mathrm{LN}\big(\mathbf{h}_{i,\tau}^{r-1} + W_O^{(r)} \mathbf{m}_{i,\tau} \big), \qquad \mathbf{h}_{i,\tau}^{r} = \mathrm{LN}\big(\tilde{\mathbf{h}}_{i,\tau} + \mathrm{FFN}^{(r)}(\tilde{\mathbf{h}}_{i,\tau}) \big). \tag{5}$$

$\mathbf{u}_{\tau-\tau'}$ is a learned lag embedding and $\mathbf{w} d_{ji}$ a distance term. Places with no upstream neighbour (Dam, T1) keep $\mathbf{h}^r = \mathbf{h}^{r-1}$. Four heads are used (written as one above for clarity).

Block 3 – Predict ahead. Each station reads its own representation at future column k:

$$\hat{\Delta}_{s,k} = g\big(\mathbf{h}_{s,k}^{4} \big), \qquad \hat{y}_{s,k} = \tilde{\ell}_{s,0} + \sigma_s\, \hat{\Delta}_{s,k}, \tag{6}$$

where g is a two-layer MLP shared by all stations (the place embedding tells them apart), $\tilde{\ell}_{s,0}$ is the latest valid level at s, and σ_s the training standard deviation of one-hour level changes at s.

Training objective. Weighted, masked mean squared error, as in the tutorial:

$$\mathcal{L} = \frac{\sum_{b,s,k} M_{b,s,k}\, w_{b,s,k} \left(\hat{y}_{b,s,k} - y_{b,s,k} \right)^2 / \sigma_s^2}{\sum_{b,s,k} M_{b,s,k}\, w_{b,s,k}}, \qquad w_{b,s,k} = 1 + \lambda \mathbb{1}\big[y_{b,s,k} \geq \theta_s \big], \tag{7}$$

with $M = 0$ for missing targets, $\lambda = 4$ (tuned on validation) and alert levels θ_s (A8). Adam, early stopping on validation error, chronological split (A1). During training, 10–20% of cells (and occasionally a whole row) are randomly masked so the model learns to forecast with missing data. *Optional extension:* g outputs the 10/50/90% quantiles, trained with pinball loss, to give a forecast range.

Why this design.

- *Upstream paths are required.* A station's own history cannot show water already on its way (companion notebook, Part 1). Block 2 gives every station a path to all upstream sources; four rounds cover the longest path, Dam→S1→S2→S3→S4.
- *Delay must be learned, not fixed.* Travel time changes with flow (F6, A10). Attention weights are recomputed from each input, so the same model can look 50 minutes back on a calm day and 25 minutes back in a flood. A fixed convolution kernel along the river would impose one delay.
- *Unequal spacing.* A spatial convolution treats index neighbours as equally related (notebook, Part 2). Here "neighbour" comes from the supplied river connectivity (F4), and distance is an explicit feature.
- *TCN for local history.* A causal dilated TCN (Lecture 4) reaches the full 36-step window with five layers and few parameters, and captures local shape (rising limb, peak) that single-value tokens lack (Lecture 5).
- *Announced releases on the same grid.* Covariates help only when they cover the forecast horizon (finding in PA1). Putting the plan in the Dam row's future cells lets it flow to S1 (and onward) through the same blocks.
- *Predicting the change.* Levels are persistent over one hour; predicting $\hat{\Delta}$ makes persistence the default, so the network learns only rises and falls.
- *Cost.* The graph is tiny (6 places, 5 edges), so the quadratic cost of attention is negligible (see Section 5).

How each source reaches each forecast.

- *Reservoir releases* (past and planned) enter at the Dam row and reach S1 in round 1, S2 in round 2, S3 in round 3 and S4 in round 4.
- *Upstream conditions*: each station attends to its upstream neighbours' histories, which already contain everything further upstream from earlier rounds.
- *Monitored tributary*: T1 is a direct upstream neighbour of S2, and so indirectly of S3 and S4.
- *Rainfall*: R1 enters at S2's row (local inflow); R2 enters at S3's row as the proxy for the unmonitored tributary (A11, A12). The TCN learns the delay between rain and river response.

Physical assumptions vs. learned mappings.

Imposed by design (physical assumptions)	Learned from data
Flow direction and connectivity: information moves only downstream (A9)	Travel time and how it changes with flow (attention over lags)
Causality: no column uses later columns; nothing unknown at t_0	Attenuation and spreading of flood waves between stations
Where each source enters the river (Dam→S1, T1→S2, rain rows; A11)	Rainfall-to-level response and its delay
Positive, bounded delay: at most 5 h (A7)	Unmonitored inflow, inferred from R2 and the S2–S3 difference
Continuity: forecast = current level + change	How reliable announced releases are (via flag s)

No separate physical simulator is used; the physics enters through the structure above. Uncertainty is addressed by the optional quantile output and by the flags described in Sections 5 and 6.

Baselines for evaluation. Persistence ($\hat{y}_{s,k} = \tilde{\ell}_{s,0}$), and an Autoformer with all places mixed into one token per step (as in PA1). Ablations: no Block 2 (each station alone), no future plan, and fixed rather than attended delays.

5. Operation and exceptional conditions

(i) Missing observations. A missing level or rain value is set to 0 with its flag (m_ℓ or m_r) set to 0, so the model can tell "missing" from "zero". Random masking during training (Section 4) teaches the model to use upstream and neighbouring rows instead. If a station's current level is missing, the forecast is anchored on its latest valid value $\tilde{\ell}_{s,0}$ and marked lower-confidence. Missing targets are excluded from the loss ($M = 0$). If a whole station has been missing for more than two hours, forecasts are still issued but flagged to operators, and the quantile range widens.

(ii) Delayed observations. The forecast is issued on time at t_0 with whatever has arrived; values not yet received are treated as missing. A reading is placed by its *observation* timestamp, but used only if its *arrival* time is $\leq t_0$. When a late reading arrives it fills its cell for the next issuance; because one forecast takes milliseconds, the current forecast can also be re-issued at once. Training simulates delays using the observed delay distribution (A4), so the model sees the same gaps it will meet in operation.

(iii) Announced future releases. The latest schedule announced by t_0 is placed in the Dam row's future columns ($\tau = 1, \ldots, 6$) by effective time, with $s = 1$. A schedule announced after t_0 is not used, even if it takes effect earlier. With no announcement, the current rate is assumed to continue (A14). Training uses the plans as they were announced at each past t_0 (A6), not the releases that later occurred.

(iv) Scheduled vs. actual releases. Once a release is recorded, its past cell holds the actual value with $s = 0$, overriding the plan. Because training pairs announced plans with the levels that followed, the model learns how far plans can be trusted, through flag s. When the actual release has departed from the plan by more than a set threshold over the last 30 minutes, operators are alerted and the forecast range is widened. Since a forecast is cheap, operators can also run "what-if" forecasts for alternative release plans.

Feasibility every 10 minutes. One forecast processes $6 \times 36 = 216$ cells. The TCN costs about $216 \times 5 \times 3 \times 64^2 \approx 1.3 \times 10^7$ multiply–adds; the four attention rounds about $4 \times 36 \times 72 \times 64 \times 2 \times 4 \approx 5 \times 10^6$; the model has roughly 2×10^5 parameters. This is well under 10 ms on a standard CPU, and data alignment is a few database queries, so a ten-minute cycle leaves a large margin. Training is done offline and repeated periodically (e.g. monthly or after a major flood).

6. Uncertainty, limitations, and additional information

Table 3: Uncertainties and limitations, their effect on forecasts, and our response.

Issue	Effect on forecasts	Response / what remains unresolved
Unmonitored tributary	Biggest blind spot. S3 and S4 may be under-forecast during storms over its catchment, especially if R2 misses a local storm.	R2 used as proxy (A12); S2–S3 difference lets the model infer recent inflow; wider ranges at S3/S4. *Unresolved* for storms not seen by R2.
Changing travel times	Timing errors: a flood peak arriving earlier than the model expects.	Flow-dependent attention over lags; high-water loss weight. *Unresolved* for floods faster than any in the record.
Floods larger than the record	Neural networks extrapolate poorly; peaks may be under-predicted.	Flag inputs outside the training range; fall back to persistence plus upstream rise. A simple physical routing model could be added later.
Future rainfall unknown	Rain falling after t_0 can raise levels within the hour, mainly at S2/S3.	Inherent at a one-hour horizon; partly shown in the range. Rain nowcasts would help (request below).
Only two rain gauges	Point readings may miss convective storms.	Learned response averages over many events; uncertainty widens in heavy rain.
Release deviations	S1 (and later S2–S4) wrong when operations depart from the plan.	Actual values override plan once recorded; deviation alert (Section 5).
Sensor faults and late data	Sudden errors or stale inputs.	Masks, random-masking training, quality screen, confidence flags.
Point forecast smooths peaks	MSE predicts a conditional mean, which flattens sharp peaks when outcomes are uncertain (notebook, Think about 1).	Quantile extension gives a range; high-water weighting reduces peak bias.
Changes over time	Channel erosion, new gauges or new reservoir rules make the past less representative.	Monitor error monthly; retrain on recent data.
Limited flood examples	Few high-water events to learn from.	Small model, shared weights across places, data augmentation by masking.

Table 4: Additional information requested from the authority.

Request	How it would help	Usable without it?
A gauge on the unmonitored tributary (even temporary)	Removes the main blind spot; a temporary record could also calibrate the R2 proxy.	Yes, with wider ranges at S3/S4.
Archived release schedules with announcement times	Lets training use plans exactly as known at each t_0 (A6).	Partly: actual releases can stand in for plans, but the model would be over-confident in plans.
Flood alert levels and historical flood reports	Sets loss weights and gives event-based evaluation.	Yes, using high percentiles of the record.
Rainfall radar or short-range rain forecasts	Adds rain expected in the next hour.	Yes; this is the current design.
Rating curves or discharge at the stations	Allows flow-based checks (e.g. water balance between stations) and estimates of the unmonitored inflow.	Yes.
Data delay logs and sensor metadata	Realistic delay simulation in training (A4) and better quality screening.	Yes, using assumed delays.
Catchment boundaries for R1, R2	Confirms which station each gauge drains to (A11).	Yes, with A11 as stated.
Reservoir level and spillway rules	Anticipates uncontrolled spills that are not in the schedule.	Yes, but spills would be missed until seen at S1.

Question 2 – Motion or Memory?

[TODO: Question 2 to be added.]

Appendix – Use of Generative AI

Claude (Anthropic) was used as an assistant. As required by the course AI policy, the main prompts, a summary of each output, and the edits I made are listed below.

Question 1

Prompt	Output (summary)	My edits / decisions
"What does part one say? And how shall I need to solve it, first help understand part 1"	Summary of the Q1 brief, the river setting and the six required headings; link between companion-notebook lessons and design choices.	[TODO: your edits]
"More cleanly we need to define an architecture choice for such a river? What are inputs, outputs?"	A first architecture (graph with lag attention) with input/output tables.	[TODO:]
"Check out things we have already in the course folder and then choose"	Review of Lectures 3–5 and PA1; a design limited to methods covered in the course.	[TODO:]
(Shared the course tutorial)	Revised design following the tutorial: one grid, look back / pass downstream / predict, 4×6 output, weighted loss.	[TODO:]
"Let's solve it now as per the brief provided and the tutorial"	Draft of Sections 1–6 and the architecture diagram.	[TODO:]

All design decisions and the final content of this report are my own.

Question 2

[TODO: log prompts, outputs and edits for Q2.]